

Ivar Cashin Eriksson

Curriculum Vitae

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Introduction

I am a curious and analytical data scientist with seven years of experience applying machine learning across healthcare, pharmaceutical analytics, manufacturing, e-commerce, and energy. My background combines KTH training in Statistical Learning and Data Analytics with hands-on work on explainable AI, representation learning, probabilistic modelling, and production ML systems. I am applying to PhD work in mechanistic interpretability and LLM security because I want to return to fundamental research on how models represent information and when their behaviour can be trusted to continue my learning journey.

Education

- 2015–2020 **Royal Institute of Technology, KTH – Data Science and Engineering**, GPA: 5.0 (out of 5.0)
Master: Statistical Learning and Data Analytics, Bachelor: Engineering Physics. [See thesis](#).
Relevant coursework: Statistical Machine Learning; Modern Methods of Statistical Learning; Reinforcement Learning.
- 2019–2019 **Swiss Federal Institute of Technology, ETH – Applied Mathematics**, *Exchange*
- 2017–2022 **Stockholm University, SU – Economics**, *Supplementary Bachelor's Degree*

Selected Work Experience

Researcher within Data Science @ RaySearch Laboratories

- Learned latent anatomical representations from segmented 3D CT scans using autoencoders.
- Predicted dose-volume histograms and spatial dose distributions with sparse Gaussian processes.
- Redesigned lexicographic optimisation and validated methods with clinicians.

Keywords: representation learning, autoencoders, sparse Gaussian processes, probabilistic modelling, optimisation, medical AI

[Read more...](#)

Data Scientist & Project Manager @ Signum Life Science

- Building an end-to-end forecasting platform for pharmaceutical price and demand forecasting.
- Owned Kwarts, a next-best-action engine for pharmaceutical sales, including explainable AI features.
- Add explainable AI features in collaboration with UX designers.

Keywords: explainable AI, AWS, Azure Databricks, forecasting, XGBoost, ML engineering

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Principal Data Scientist @ Valcon

- Built a production real-time neural network using natural-language inputs and high-dimensional labels.
- Held explainable AI and Git trainings; mentored junior data scientists.
- Presented complex technical topics to non-technical stakeholders.

Keywords: natural-language inputs, deep neural networks, explainable AI, MLOps, mentoring

[Read more...](#)

Skills and Other

- Technology Python (PyTorch, TensorFlow, SHAP, YOLO), C++, Matlab, MS Azure (Data Science Associate Certified), Databricks, AWS, MongoDB, Qdrant, Docker.
- Methods Explainable AI, autoencoders, CNNs, sparse Gaussian processes, clustering, vector search.
- Language *Native:* English, Swedish. *Intermediate:* Danish.
- Other Member of Mensa Sweden.

Detailed Work Experience

2024–Present **Full-Stack Data Scientist and Project Manager**, *Signum Life Science*, Copenhagen

At Signum, my current work is on an end-to-end forecasting platform for pharmaceutical price and demand forecasting. The work builds on a pharmaceutical price forecasting model that helps clients navigate Denmark's fortnightly blind auctions. After stabilising the XGBoost-based system, I am now driving efforts toward a more robust and modern implementation.

Earlier in the role, I owned and extended Kwarts, a next-best-action engine for pharmaceutical sales. Originally built externally, the system was extended with features such as explainable AI in collaboration with UX designers, and supported commercialisation by aiding our sales representatives. Kwarts adapts to varied customer data maturity, from rule-based logic to deep learning, and is deployed in AWS for scalability.

Beyond model development, I organised an internal week-long hackathon that included up to 40 participants.

2022–2024 **Principal Data Scientist**, *Valcon*, Copenhagen

Led data science projects across pharma, energy, IoT, and manufacturing. Mentored junior colleagues in software architecture and data science best practices, led explainable AI and Git trainings, and regularly presented complex technical topics to non-technical stakeholders.

One project involved training a real-time deep neural network with a custom training pipeline and modular architecture. The training data consisted of natural-language inputs and high-dimensional multi-class labels, which made for a challenging prediction problem. I was the lead data scientist and architect of the solution and responsible for implementing the end-to-end production pipeline.

I was awarded the "Valcon People's Choice Award" for exemplary work, leadership, team spirit, and scientific curiosity.

2021–2022 **Data Scientist**, *Valtech*, Copenhagen

Delivered ML solutions in e-commerce. Built time series forecasts using Facebook Prophet, and applied clustering to behavioural data to support personalisation. Worked closely with business stakeholders to ensure actionable outputs.

2019–2021 **Researcher within Data Science**, *RaySearch Laboratories*, Stockholm

At RaySearch Laboratories, I worked on automating radiotherapy planning using a combination of deep learning, probabilistic modelling, and multi-objective optimisation. I developed a pipeline that extracted latent anatomical features from segmented 3D CT scans using autoencoders, and used these representations to identify similar patients via a learned similarity metric. Dose-volume histograms and spatial dose distributions were then predicted using sparse Gaussian processes, providing a probabilistic framework for dose planning conditioned on patient geometry.

Alongside this, I redesigned some of RaySearch's lexicographic optimisation algorithms to better reflect clinical decision hierarchies, particularly in palliative care where trade-offs between tumour control and organ sparing are nuanced. I collaborated closely with medical physicists and clinicians throughout, ensuring the models aligned with oncological standards and practical workflow requirements.

Other Detailed Experience

2025–2025 **Solo Project**, *Project Iris*

In this personal project I built a PoC end-to-end computer vision system for making product images on e-commerce sites interactive. The pipeline detects individual items using a YOLOv8 model, and embeds them semantically using OpenCLIP. The embeddings are then compared to support intelligent product linking across all images and products in a web store.

The system includes a full-stack setup with a FastAPI backend, MongoDB for persistence, a Qdrant index for vector search, and a custom-built Chromium plug-in that overlays clickable links directly onto web shop imagery in real time. The project also includes a dynamic scraping and indexing pipeline for ingesting and structuring product data from arbitrary e-commerce sites.

The project was a learning opportunity for fine-tuning of deep learning models and practical ML engineering.